

Process excellence and capability determination

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BT's Systems Engineering department is faced with a constant challenge to deliver quality software against increasing customer demands. It is important to understand the processes BT uses and to be able to continually improve them. The first step was the adoption of ISO9001 [1] and then ISO9000-3 (TickIT) [2]) within BT. These standards were, however, while useful, considered too prescriptive to fully meet our needs. A new standard, ISO 15504 also known as SPICE, moves us forward from the prescriptive audit method towards a flexible assessment approach. SPICE provides a measure of the maturity of a process against a defined model of best practice, rather than a pass/fail result as with ISO9001. This paper presents an overview of the emerging SPICE standard, together with a discussion on its potential benefits for Systems Engineering and BT.

1. Introduction

BT's Systems Integration department, part of Systems Engineering (SE), strives continually to improve its performance as a business. The emergence of the 'trading unit' philosophy now brings the greater challenge of becoming even more cost-conscious, schedule-conscious and customer-conscious. This mandates real and continuous change for all developers, integrators, and acquirers of systems.

The issues of time and quality have been with the industry for many years now. At a superficial level they are little different to the issues faced by more traditional manufacturing industries, where much of the real improvement has come from the enhanced understanding of procedures or 'processes' which has enabled the definition of streamlined standards against which to work.

This has enabled the measurement of the success, or otherwise, of a particular process in meeting its defined goals. Consequently, methods have been devised to analyse the performance of the process, and to manipulate (process re-engineer) it such that these new standards are met.

In the relatively immature IT industry, research into the processes which constitute good software development practices only started in the mid-1980s. As a result, it has taken until the 1990s for the relevant and appropriate standards to emerge. Perhaps the best known of these is ISO9000 supported by TickIT (its interpretation for software developers) [2]. ISO9000 and TickIT provide a definition of procedures against which an organisation is compared through an audit. The main criterion is to find areas of non-compliance with the standard. This is done by checking for

the existence of evidence that procedures have been carried out as defined within the quality management system.

More recently, research has focused upon measurement of the capability of the individual process to perform the task for which it is defined. This calls for a move away from the audit approach to one of expert assessment of the attributes that characterise the process. This approach allows an assessor to rate the attributes of the process, not just in terms of whether the process is performed and complies, but also as to how it contributes to the goals of the project and organisation. Is the process well defined and understood by those who use it? Does it offer improvement opportunity? This type of assessment, known as process assessment, can be used in three ways:

- to assess the processes employed by an organisation as part of a performance improvement programme,
- to assess the processes employed by an organisation to identify areas of risk in that organisation's capability to deliver, normally undertaken as part of a supplier selection exercise,
- to assess the organisation's processes against other organisations.

Much of the research into IT processes has been carried out by an international team known as the SPICE project, which has been supported by a working group of the International Standards Organisation (ISO/IEC/JTC1/SC7/WG10). SPICE is an acronym derived from Software Process Improvement and Capability dEtermination.

2. Assessment, improvement and capability

Process assessment is the systematic analysis of processes to provide a report and action plan either for improvement or as part of a supplier selection exercise. Assessments are conducted mainly through interviews using competent assessors who deliver a formal report and action plan. While differing slightly in detail, assessments for improvement purposes and assessments for supplier management can be defined as having the following major steps:

- initiation — defining the need for the assessment and its purpose,
- preparation — plan the assessment, set targets and communicate the approach to all affected people,
- assessment — capture the data and review relevant supporting evidence,
- analysis/reporting — analyse the data and inform the relevant personnel of findings,
- closure — examine the assessment performance against the assessment purpose.

Assessment at its simplest is a measurement tool. It can be compared to the dashboard in a car where the instruments help you drive but they do not do the driving for you. Equally each specific assessment yields significant data, but it is the analysis of the information and the subsequent actions that produce results. The analysis that follows assessment therefore turns the raw data into a report and action plan. The power of assessment lies in the quality of the information it provides to form the basis for the analysis.

A major feature of assessment is the ability to set focused targets for performance and to analyse any gaps between the target and actual performance (see section 5).

Process assessment has two principal contexts for its use, as shown in Fig 1.

- In a process improvement context, process assessment provides the means of characterising the current practice within an organisational unit in terms of the capability of the selected processes. Analysis of the results in the light of the organisation's business needs identifies strengths, weakness and risks inherent in the processes. This, in turn, leads to the ability to determine whether the processes are effective in achieving their goals, and to identify significant causes of poor quality, or overruns in time or cost. These provide the drivers for prioritising improvements to processes.

- Process capability determination is concerned with analysing the proposed capability of selected processes against a target process capability profile in order to identify the risks involved in undertaking a project using the selected processes. The proposed capability may be based on the results of relevant previous process assessments, or may be based on an assessment carried out for the purpose of establishing the proposed capability. (Capability determination is normally undertaken as part of a supplier selection exercise.)

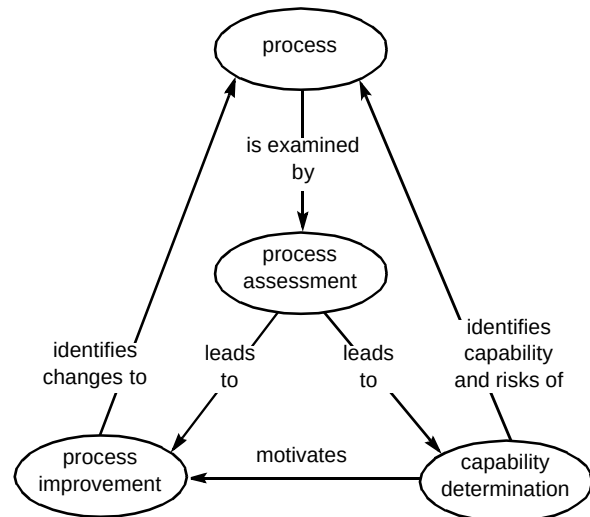


Fig 1 Software process assessment.

2.1 Using process assessment for improvement

Implementing sustainable improvement requires people to recognise opportunities for improvement and change their behaviour to match. Improvement is therefore about managing, building and maintaining a culture for change. Process assessment can be used as part of process management programmes for any of eight purposes:

- skill transfer — internal benchmarking of a successful project can guide others to adopt similar approaches for similar or appropriate projects,
- unlocking opportunity — assessment allows people quality time away from current pressures to consider the details of how they can work in smarter ways,
- motivation for change — assessment actively encourages people (participants and assessors) to own the improvements captured,
- pinpointing improvement — assessment is systematic in its probing of activities giving a hitherto unattainable completeness in identifying improvement opportunities,

- exploring needs — the underlying SPICE model (see section 4) allows processes to be explored not only for improvement purposes but also to build new approaches to match new ventures,
- analysing improvement — repeated assessments provide trend information that discloses which improvements provide impact and the scale of such impact,
- benchmarking capability — comparing local processes with those of other major players in the software and telecommunications industries will allow strengths to be consolidated and weaknesses tackled in a more focused way,
- continuous improvement — continuous improvement means that an organisation's workforce is continuously motivated to improve on current good practice and manage changes in a measured way.

2.2 Identifying improvement opportunities

Improvement opportunities are captured firstly from participants and then by analysing the full set of results from the assessment. Here the focus is on the internal operation of individual processes. Further improvement opportunities that occur across processes are also exposed due to the structure of the underlying process model.

An important feature in obtaining information is the confidentiality of results. For improvement data capture it is important that participants neither feel intimidated nor are concerned that their individual comments will be used in evidence against them. Here a clear definition of which parts of the assessment will be available to specific levels of the organisation has to be carefully set, communicated and observed.

2.3 Identifying strengths and weaknesses

The assessment method is specific in capturing the overall process strengths and weaknesses from the people interviewed and the documentation reviewed. This is valuable in gauging the priority of the improvements from the participants and assists in the production of the assessment report and the subsequent preparation of the action plan.

2.4 Process assessment and capability determination

Capability determination is primarily used for supplier management. The objectives of supplier management are:

- to provide a basis for intelligent comparison of suppliers and their approaches,

- to tailor both parties' processes to best achieve a project's goals,
- to implement a closer working relationship between suppliers and acquirers,
- to quantify the risks associated with using specific suppliers.

Assessment represents a major innovation in how suppliers and acquirers can work together. It forces a more open discussion of the capability expected and the approach to be used than has previously been typical. This can considerably enhance negotiations in the early stages of projects to ensure a solid understanding of each party's responsibilities and expectations.

2.5 Setting selection criteria and benchmarks

The model in use within SE [3] allows a target capability to be set against processes that it is believed are necessary to manage a specific type of project or produce a specific type of product. This target may then be used to assess the capability of specific suppliers for individual projects.

3. Background to the SPICE project

The SPICE project is an international collaborative effort to support the development of a new international standard for software process assessment. Its objective is to create a standard for the assessment of software processes, bringing together the best points from all existing methods such as the Software Engineering Institute's capability maturity model [4], BT's Healthcheck method [5], Trillium from Bell Northern [6], Bootstrap [7], and the Software Technology Diagnostic [8]. The Systems Integration department has been involved in, has supported and has been committed to the emerging international standard since its inception.

The project was established by the international committee on software engineering standards, ISO/IEC JTC1/SC7, through its Working Group 10 on software process assessment, in January 1993. This followed the recommendations of the committee's study group on the needs and requirements for a standard for software process assessment, which in 1992 recommended the following approach [9]:

- that development should be carried out by a dedicated project team,
- that development should be closely linked with a period of field trials and experimentation,

- that the standard should be published initially as a technical report (Type 2) to enable the method to stabilise and user feedback to be obtained prior to committing to a full standard.

Development by a dedicated team, with a core team staffed by full-time resource, will provide greater scope for more intensive technical design and development and should result in a reduction in time-scales.

Following on from acceptance of the study group report, the task of defining a standard was assigned to a new Working Group, WG10, which established the SPICE project to support this goal [10–18]. The origins of the project are described in more detail by Dorling [19], together with the management structure adopted to direct the project.

The project comprises a core team staffed by full-time resource, with the goal of providing greater scope for more intensive technical design and development. Four international technical centres were established to co-ordinate the international effort on the project. Its conduct has brought together a unique combination of standards developers, software developers and academics from leading companies and institutions (including BT's Systems Integration department) from 20 countries, to generate a consistent and validated framework for assessment in a remarkably short period of time. The success of the approach can be gauged from the results — while the project has fallen behind its initial schedule, it produced a full suite of documents suitable for extensive industry trials in a period of two years.

4. SPICE structure, use and benefits

SPICE defines a reference model of processes and process capability [11] that forms the basis for any model to be used for the purposes of process assessment. This reference model comprises a two-dimensional approach to the evaluation of process capability — one dimension defines the processes to be assessed, the other describes the scale for measurement of capability. Any model [14] compatible with the reference model may be used for assessment, and the results of any conformant assessments will be able to be translated into a common base, thus allowing direct comparisons, for example, for benchmarking.

4.1 The process dimension

The reference model groups the processes in the process dimension into five process categories, according to the type of activity they address:

- the customer/supplier process category consists of processes that directly impact the customer, support development and transition of the software to the customer, and provide for its correct operation and use,
- the engineering process category consists of processes that directly specify, implement, or maintain a system and software product and its user documentation,
- the support process category consists of processes that may be employed by any of the other processes (including other supporting processes) at various points in the software life cycle,
- the management process category consists of processes that contain practices of a generic nature that may be used by anyone who manages any sort of project within a software life cycle,
- the organisation process category consists of processes that establish the business goals of the organisation and develop process, product, and resource assets which, when used by the projects in the organisation, will help the organisation achieve its business goals.

Process categories and processes provide a grouping by type of activity. Each process in the reference model is described in terms of a purpose statement. For example, the engineering process category contains the processes:

- ENG.1 develop system requirements and design,
- ENG.2 develop software requirements,
- ENG.3 develop software design,
- ENG.4 implement software design,
- ENG.5 integrate and test software,
- ENG.6 integrate and test system,
- ENG.7 maintain system and software.

Picking one process from the engineering process category, the purpose of the process 'integrate and test software' is as follows [11]:

"The purpose of the integrate and test software process is to integrate the software units with each other, producing software that will satisfy the software requirements. This process is accomplished step by step by individuals or teams. As a result of successful implementation of the process an integration strategy will be developed for software units, consistent with the release strategy:

- acceptance criteria for aggregates will be developed that verify compliance with the software requirements allocated to the units,

- software aggregates will be verified using the defined acceptance criteria,
- integrated software will be verified using the defined acceptance criteria,
- test results will be recorded,
- a regression strategy will be developed for re-testing aggregates or the integrated software should a change in components be made.”

These statements comprise the unique functional objectives of the process when instantiated in a particular environment. Satisfying the purpose statements of a process represents the first step in building process capability.

The reference model does not define how, or in what order, the elements of the process purpose statements are to be achieved. The process purposes will be achieved in an organisation through the various lower level activities, tasks and practices being carried out to produce work products. These performed tasks, activities and practices, and the characteristics of the work products produced, are the indicators that demonstrate whether the specific process purpose is being achieved.

Evolving process capability is expressed in terms of process attributes grouped into capability levels. Process attributes are features of a process that can be evaluated on a scale of achievement, providing a measure of the capability of the process. They are applicable to all processes. Each process attribute describes a facet of the overall capability of managing and improving the effectiveness of a process in achieving its purpose and contributing to the business goals of the organisation.

A capability level is a set of attribute(s) that work together to provide a major enhancement in the capability to perform a process. Each level provides a major enhancement of capability in the performance of a process. The levels constitute a rational way of progressing through improvement of the capability of any process. The attributes are allocated to the capability levels as shown in Table 1.

Table 1 Allocation of attributes to capability levels.

Scale	Process attributes
Level 1	Process performance
Level 2	Performance management Work product management
Level 3	Process definition and tailoring Process resource
Level 4	Process measurement Process control
Level 5	Process change Continuous improvement

4.2 The capability dimension

There are six capability levels in the reference model, incorporating the nine process attributes above. The levels are defined as follows.

- **Level 0.** Incomplete — there is general failure to attain the purpose of the process. There are no easily identifiable work products or outputs of the process.
- **Level 1.** Performed — the purpose of the process is generally achieved. The achievement may not be rigorously planned and tracked. Individuals within the organisation recognise that an action should be performed, and there is general agreement that this action is performed as and when required. There are identifiable work products for the process, and these testify to the achievement of the purpose.
- **Level 2.** Managed — the process delivers work products of acceptable quality within defined time-scales. Performance according to specified procedures is planned and tracked. Work products conform to specified standards and requirements. The primary distinction from Level 1 (Performed) is that the performance of the process is planned and managed and progressing towards a defined process.
- **Level 3.** Established — the process is performed and managed using a defined process based upon good software engineering principles. Individual implementations of the process use approved, tailored versions of standard, and documented processes. The resources necessary to establish the process definition are also in place. The primary distinction from Level 2 (Managed) is that the process is planned and managed using a standard process.
- **Level 4.** Predictable — the defined process is performed consistently in practice within defined control limits, to achieve its goals. Detailed measures of performance are collected and analysed. This leads to a quantitative understanding of process capability and an improved ability to predict performance. Performance is objectively managed. The quality of work products is quantitatively known. The primary distinction from Level 3 (Established) is that the defined process is quantitatively understood and controlled.
- **Level 5.** Optimising — performance of the process is optimised to meet current and future business needs, and the process achieves repeatability in meeting its defined business goals. Quantitative process effectiveness and efficiency goals (targets) for performance are established, based on the business goals of the organisation. Continuous process monitoring against these goals is enabled by obtaining quantitative feedback and improvement is achieved by

analysis of the results. Optimising a process involves piloting innovative ideas and technologies and changing non-effective processes to meet defined goals or objectives. The primary distinction from Level 4 (Predictable) is that the defined process and the standard process undergo continuous refinement and improvement, based on a quantitative understanding of the impact of changes to these processes.

The reference model alone cannot be used as the basis for conducting reliable and consistent assessments of process capability since the level of detail is not sufficient. The descriptions of process purpose and capability attributes in the reference model need to be supported with a comprehensive set of indicators of process performance and capability [14]. In this way, consistent ratings of process capability will be possible.

4.3 Rating process attributes

A process attribute represents a measurable characteristic of any process as defined above. The rating scale defined in Table 2 is used to describe the levels of achievement of the defined capability of the process attributes.

Table 2 Levels of achievement.

N	not achieved — there is no evidence of achievement of the defined attribute
P	partially achieved — there is some achievement of the defined attribute
L	largely achieved — there is significant achievement of the defined attribute
F	fully achieved — there is full achievement of the defined attribute

4.4 Benefits of the SPICE approach

The major benefits of a standardised approach to process assessment are that it will:

- provide a public, shared model for process assessment,
- lead to a common understanding of the use of process assessment for process improvement and capability evaluation,
- facilitate capability evaluation in open procurement,
- be controlled and regularly reviewed in the light of experience of use,
- be changed only by international consensus,
- encourage harmonisation of existing schemes [10].

The approach to process assessment defined in SPICE is designed to provide a basis for a common approach to describing the results of process assessment, allowing for some degree of comparison of assessments based upon different, but compatible, models and methods. The sophistication and complexity required of a process is dependent upon its context. For instance the planning required for a five-person project team is much less than for a fifty person team. This context influences how a qualified assessor judges a practice when assessing its adequacy and influences the degree of comparability between process profiles. This comparability of assessments based on different but compatible models means that BT's Systems Integration department can compare its performance as measured by the PPP model against other companies, world-wide. Various models may have been used such as PPP, Bootstrap, the SEI capability maturity model (CMM), or other models that may emerge in the future. There are also opportunities for comparing assessment results with Joint Venture or sub-contract companies to identify differences in capability which may lead to synergy or pose a risk to the success of the undertaking.

5. Assessment within BT

Systems Integration department has, over the past year, been conducting SPICE assessments using PPP, a SPICE-compliant assessment method developed by Compita Ltd with input from the UK Process Assessment Consortium (UPAC), of which Systems Integration department is a founder member. Other units, including BT's Belfast and Ipswich software engineering centres, have also been involved in SPICE assessment. The approach follows the steps described in the next three sub-sections.

5.1 Assessing the right thing

The unit's (or programme's) key business drivers are determined either from a business or quality plan or by discussion with the unit's (or programme's) management team. For example, a unit may have as its mission statement 'to deliver systems on time, at the specified quality and at a reasonable cost to the business'. After the key business drivers have been identified, they are ranked for importance and weighted by the management team. In the example above, they may decide that timely delivery is of most importance, and quality and cost are secondary.

The processes in the model are then assessed for their contribution to the business drivers, and weighted in terms of their contribution. The contribution weighting is then multiplied by the business driver weighting, which gives a ranked list of processes in terms of their importance to the attainment of the unit's business goals. The highest ranked processes are the ones which contribute most, and therefore will deliver the most significant business benefit per improvement pound.

5.2 Setting targets

The assessor and management team then establish target ratings for the processes to be assessed, to represent Systems Integration department's ideal process capability for each of the processes, i.e. 'where we would like to be'. For example, the target for the performance management attribute should be 'fully achieved' for processes which strongly contribute to a business driver associated with achieving time-scales. The target profile of such a process might look like that depicted in Table 3.

Table 3 A sample target profile.

Scale	Process attributes	Target rating
Level 1	Process performance	Fully achieved
Level 2	Performance management	Fully achieved
	Work product management	Fully achieved
Level 3	Process definition and tailoring	Largely achieved
	Process resource	Largely achieved
Level 4	Process resource	Largely achieved
	Process measurement	Partially Achieved
	Process control	Partially Achieved
Level 5	Process change	Not Rated
	Continuous improvement	Not Rated

The process attributes of the PPP model map to those of the SPICE standard.

5.3 Results and analysis

The selected processes are assessed [12], usually by interviewing the practitioners, and a report prepared. The report details the assessment findings in terms of an assessment profile, a measurement of the level of attainment of each process against the process attributes, the process attributes representing the measurable characteristics of the process being assessed. This in turn leads to a gap analysis by comparison of the actual profile (i.e. where we are) with the target (i.e. where we want to be).

A SWOR (strengths, opportunities, weaknesses and risks) analysis is done using the results of the gap analysis, with strengths identified where processes meet or exceed their targets, and weaknesses where they fail to meet the targets. The weaknesses are rated in terms of business risk (minor, significant or substantial risk). For example, if a weakness was found in the testing process which was ranked as substantial, the actual risk may be a three-month delay on the delivery date.

Improvement opportunities are identified both during assessment interviews and the analysis of the full set of results from the assessment. These are presented in a comprehensive report to enable the assessment sponsor to produce an improvement action plan [16].

6. Conclusions

SE has used the SPICE standard to assess the capability of units and programmes covering approximately one fifth of its development effort in the last year and a half, with more assessments planned for the future. The assessment reports have been used to generate improvement plans alongside the more technically detailed 'Healthcheck' [5] assessment results delivered to all units in SE annually.

Another use of the SPICE standard, using the Process Professional [3, 20] tool-set has been to benchmark SE and specifically the network models and support unit within Systems Integration department with its counterpart in the MCI organisation. The SPICE process model provided a framework for people from both organisations to use in the analysis of their main processes, identified by the SPICE process selection exercise, in order to identify their relative strengths and weaknesses. This enabled them to identify where the two organisations are able to learn from each other, as well as highlighting risks in their own processes that might lead to failure.

The results of the BT part of the SPICE assessment are shown in Fig 2.

The assessment chart (Fig 2) shows the five levels, subdivided into one or more attributes, for ease of pinpointing problems (e.g. Level 1 has only one attribute, while Level 3 has four).

At the start of the assessment, for each process to be assessed, a target was set for the process, representing the level of attainment the assessment team judged was appropriate for that process, given BT's background and knowledge of levels of performance in the industry as a whole. The targets are shown above the actual assessment results, distinguished by dotted lines, for ease of reference.

Each attribute in each level was assessed for each process. The attributes were then marked on the four-point scale — Fully (the unit was fully meeting the process purpose for that attribute) through Largely and Partially to Not (no evidence of any capability whatsoever).

While the unit met and, in most cases, exceeded its target profile, improvement opportunities can still be

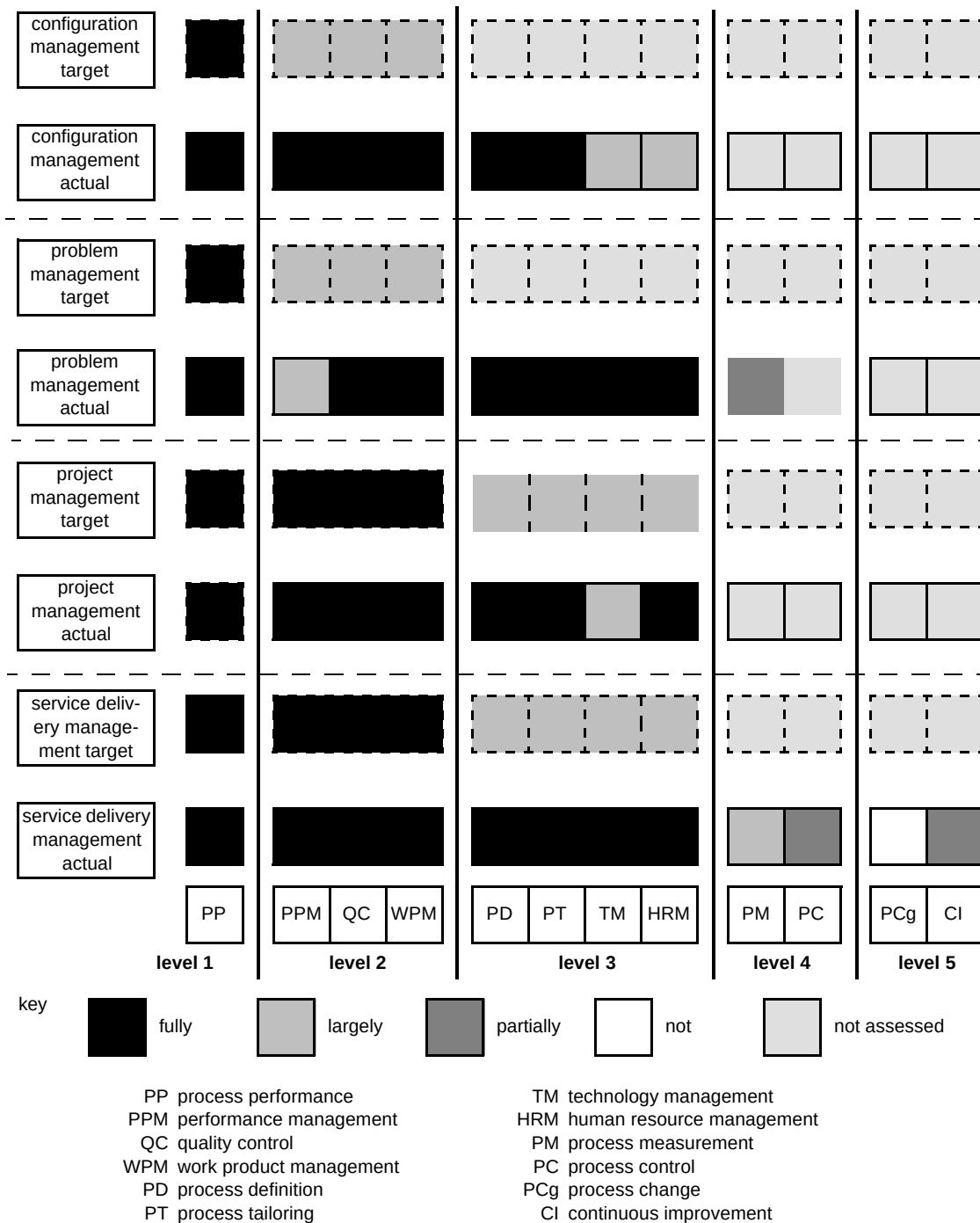


Fig 2 BT SPICE assessment.

identified to enhance the unit's capability. Every rating short of fully (i.e. every square with lighter tints) is supported by a commentary from the assessor specifying exactly the shortfall and, where appropriate, suggesting the improvement action that would eliminate it.

In the last 20 years or so there has been strong interest in, and debate about, process improvement and its real potential. Fortunately there is now substantial published

proof available that focused process improvement delivers results. For example, many organisations in the IT software industry who have invested in process improvement have reported a return in excess of five times their investment when measured over an extended period.

Table 4 contains data from 360 organisations within the IT software industry from 15 European countries, taken from a recent survey undertaken by IBM [21]. It illustrates

the gains in several areas between organisations with high business performance (in terms of development productivity and defect removal) compared with those with low performance.

Table 4 Business performance comparison.

Benefit area	Gain of high business performance organisations over low performance organisations
Productivity	at least 5 times better
Maintenance efficiency	at least 10 times better
Product quality	over 30 times better
Achievement of schedules and cost	4 times better

Systems Integration department is in a position to be able to accurately measure its process through SPICE compliant methods. The data gathered from the assessment exercise can then be used to drive process improvement, leading to gains in productivity, product quality, and delivery time-scales. This, together with the other process excellence initiatives, such as investors in people and European Quality Award self-assessment, will ensure that Systems Integration department and Systems Engineering will be in a strong position to meet any challenges the future business environment may bring. In turn, the gains achieved will ensure that BT's service to its customers is truly world class.

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